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Research on Multi - Task - Driven Multi - Organ Medical Image Segmentation and Classification Model

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Abstract. This study proposes a unified multi-task model for simultaneous multi-organ medical image segmentation and classification, aiming to enhance efficiency and versatility in clinical ultrasound image analysis. The model integrates a segmentation module based on nnUNet, which ensures high-precision organ contouring through adaptive network configuration and a Dice-CrossEntropy joint loss function, and an improved multi-head EfficientNet for classification, enabling dynamic adaptation to different organ tasks via root filename parsing. A hierarchical loss strategy with dynamic weight adjustment balances segmentation and classification performance. Experimental results on the UUSIC dataset demonstrate favorable performance: the segmentation module achieves average Dice coefficients ranging from 0.721 (thyroid) to 0.948 (fetal head), while the classification module yields accuracies between 0.507 (appendix) and 0.591 (breast lesions). This lightweight, single-model framework simplifies deployment and adapts flexibly to diverse clinical needs, though further optimization is required to address feature competition in heterogeneous tasks.

Keywords: Multi-task learning; medical image segmentation; image classification; ultrasound imaging; nnUNet; EfficientNet

1 Introduction

In the field of biomedical engineering, accurate and efficient analysis of ultrasound images of multiple organs is crucial for clinical diagnosis. The Universal Ultrasound Image Challenge (UUSIC) provides a practical platform for advancing the development of multi - organ classification and segmentation technologies using ultrasound data, driving algorithms to better meet clinical requirements. This model aims to develop a unified model that can complete the classification and segmentation tasks of multiple organs. While drawing on the advantages of existing models, it optimizes efficiency and versatility.

This model constructs a lightweight system that integrates multi-organ segmentation and classification. It uses the classic nnUNet as the core segmentation model to ensure high - precision organ segmentation. The classification module is based on EfficientNet and improved into a multi - head CNN (MultiTaskEfficientNet).

An innovative approach is proposed to intelligently identify task scenarios by parsing the root file names of data and dynamically adapt the classification heads, realizing a "single-model multi-task" operation mode. This significantly simplifies the deployment process and reduces the dependence on computing power resources, providing a more efficient and flexible technical method for clinical diagnosis and medical research.

2 Methods

2.1 Multi - task Paradigm

This project adopts the multi - task learning paradigm, aiming to solve the classification and segmentation tasks of different organs simultaneously, forming an integrated "segmentation-classification" multi-task processing flow to meet the needs of clinical image analysis. In this project, the classification task is to determine the type of organs in the ultrasound image, and the segmentation task is to accurately outline the boundaries of the organs. At the same time, by sharing computing resources and feature representations among multiple tasks, the overall efficiency and versatility of the model can be improved.

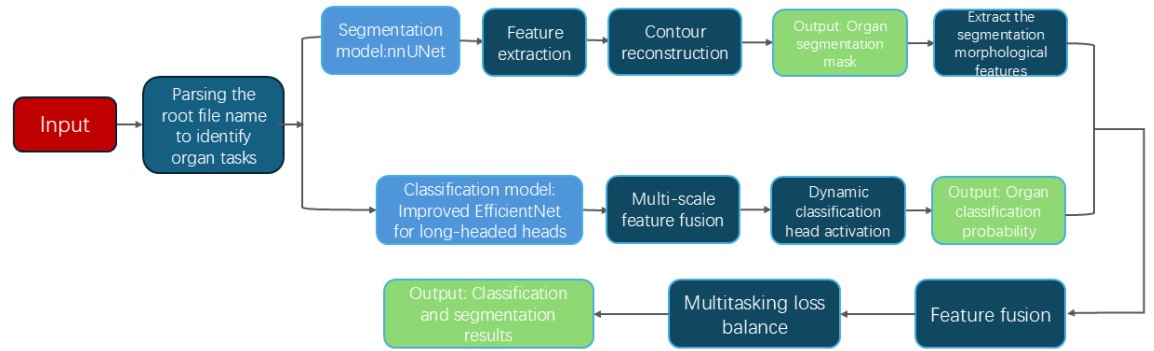


Fig. 1. Flowchart of the proposed model. It includes components like "Segmentation model: nnUNet" (blue box), "Classification model: Improved EfficientNet for long - headed heads" (blue box), and "Output: Classification and segmentation results" (green box), illustrating the model's processing steps from input to final outputs.

2.2 Model Architecture

2.2.1 Segmentation Model (nnUNet)

Traditional medical image segmentation relies on hand - designed regularization methods, such as edge detection techniques, K-clustering, region growing methods, threshold segmentation, etc.^[1], These methods rely on prior rules and have certain limitations, making it difficult to adapt to the complex variability of medical images. This study constructs a segmentation module based on nnUNet, whose core advantage lies in its adaptability and robustness to multi-organ tasks^[2]. It is used for the segmentation and classification of ultrasound images of multiple organs. This framework consists of three main parts: a feature encoding - decoding network based on the U - shaped architecture, an automatic adaptation and optimization module, and an inference guarantee module. Research results show that the multi - task network framework achieves better results on multi-organ medical image sets compared to single - task models^[3].

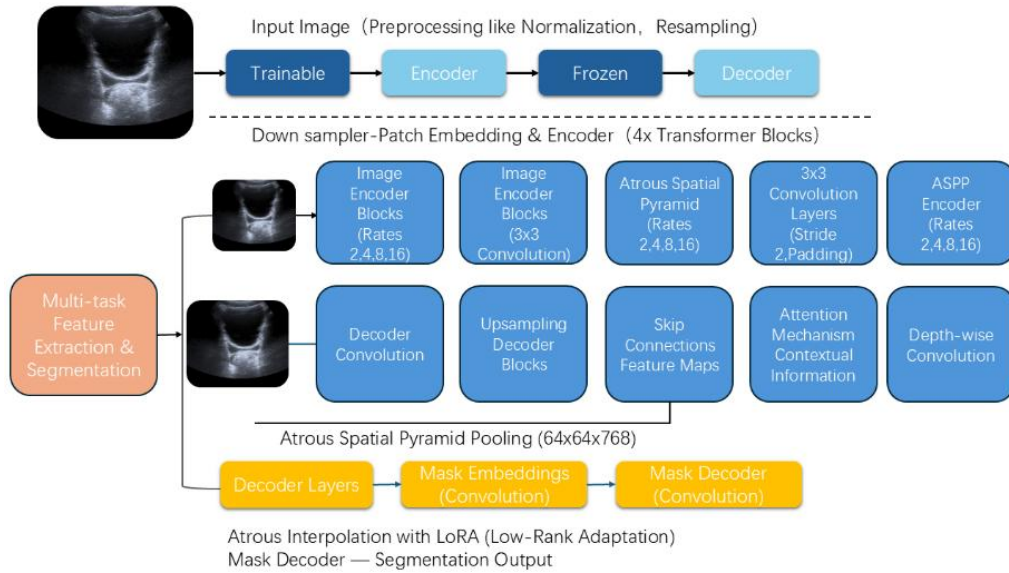


Fig. 2. Flowchart of the multi - task medical image processing model. It includes components like “Input Image (Preprocessing like Normalization, Resampling)” (top - left), “Multi - task Feature Extraction & Segmentation” (orange block), and “Decoder Layers→Mask Embeddings→Mask Decoder” (yellow blocks), illustrating steps from image input to segmentation output with atrous interpolation and transformer - related operations.

(1) Feature Decoupling and Reconstruction Based on the U-shaped Architecture

nnUNet inherits the encoder - decoder paradigm of the classic UNet, but adapts to medical images through dynamic network topology optimization^{[4][5]}. Its core consists of an encoder and a decoder. The encoder (Encoder) uses nested

downsampling (max - pooling + convolution) to gradually compress the spatial dimension and expand the number of channels. The mathematical expression is:

$$F_l = \text{Downsample}(\text{Conv}(F_{l-1}))$$

Where $F_l \in \mathbb{R}^{H_l \times W_l \times C_l}$ is the feature of the l-th layer (H_l , W_l are the resolution, and C_l is the number of channels) , Through 5 - level downsampling, the resolution of the feature map is reduced from the input $H_0 \times W_0$ to $H_5 \times W_5 \approx (H_0/32) \times (W_0/32)$, and the number of channels is expanded from C_0 to $C_5 = 32 \times 2^5$, realizing hierarchical feature extraction from local textures (F_1) to global semantics (F_5) The decoder (Decoder) restores the spatial dimension through upsampling (transposed convolution + convolution) and fuses the skip - connection features of the corresponding levels of the encoder. The formula is:

$$\hat{F}_l = \text{Upsample}(\text{Conv}(F_{l+1})) \oplus \text{Conv}(F_l^{\text{skip}})$$

where $\oplus$ represents feature concatenation (fusion in the channel dimension), enabling the decoder to utilize high - level semantics (F_{l+1}) ,and supplement low - level details (F_l^{skip}) ,to accurately restore organ contours (such as liver blood vessel branches and mammary ducts).

(2)Automatic Adaptation and Optimization Module

The core advantage of nnUNet lies in its fully automated process without the need for manual parameter adjustment, which can specifically adapt to the characteristics of medical images and achieve efficient segmentation tasks. Its operation relies on the following key mechanisms:

In terms of adaptive network configuration, the model automatically selects the network topology (including 2D/3D U - Net, cascaded U - Net, etc.) based on the statistical characteristics of the input data. Facing scenarios mainly composed of low - resolution 2D ultrasound images, it defaults to calling 2D U - Net and flexibly adjusts the patch size (such as setting it to 128×128) to balance GPU memory usage and feature utilization, enabling the model to reach an optimal state between hardware adaptation and feature extraction.

For the common problem of class imbalance in medical images, nnUNet uses the Dice - CrossEntropy joint loss function, and the formula is:

$$\zeta_{seg} = \alpha \cdot \zeta_{Dice} + (1 - \alpha) \cdot \zeta_{CE}$$

where $\zeta_{Dice} = 1 - \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|}$ (focusing on the regional overlap degree) , $\zeta_{CE} = - \sum Y \log(\hat{Y})$ (strengthening pixel classification) , and $\alpha \in [0,1]$ is the balance coefficient. This loss function has stronger robustness to the extremely unbalanced problem of "foreground (organ) - background (non - organ)" in medical images (such as the segmentation of small lesions).

In addition, in the data augmentation and generalization stage, the model has built - in strategies such as elastic deformation, random rotation, and intensity scaling

to simulate clinical acquisition differences (such as the imaging noise of different ultrasound probes). Mathematically, it is expressed as:

$$I_{aug} = Elastic(I) \circ Rotate(I) \circ Scale(I)$$

Where $\circ$ represents the transformation combination, enabling the model to learn more generalized features and adapt to medical image data from multiple centers and devices.

(3) Inference Guarantee

In the inference stage, nnUNet uses a sliding window to process large - sized images. The segmentation results can also assist the classification task through spatial constraints. Morphological features of the segmentation mask are extracted to construct a feature vector f_{seg} , and f_{seg} is fused with the image features of the classification model to enhance the multi - task collaboration performance.

2.2.2 Classification Model (Improved Multi - head EfficientNet)

In traditional deep - learning methods, most models design independent learning networks for single tasks^[6]. However, although there are specificities in the classification tasks of different organs (such as differences in morphological features and lesion types), there are internal correlations in their underlying image features (such as gray - level distribution and boundaries). Therefore, this classification model constructs a multi - task learning framework based on EfficientNet and realizes the collaborative optimization of multi - organ classification tasks by improving it into a multi - head architecture. It mainly consists of three modules: task recognition, multi - head structure and multi - scale feature fusion, and efficiency improvement.

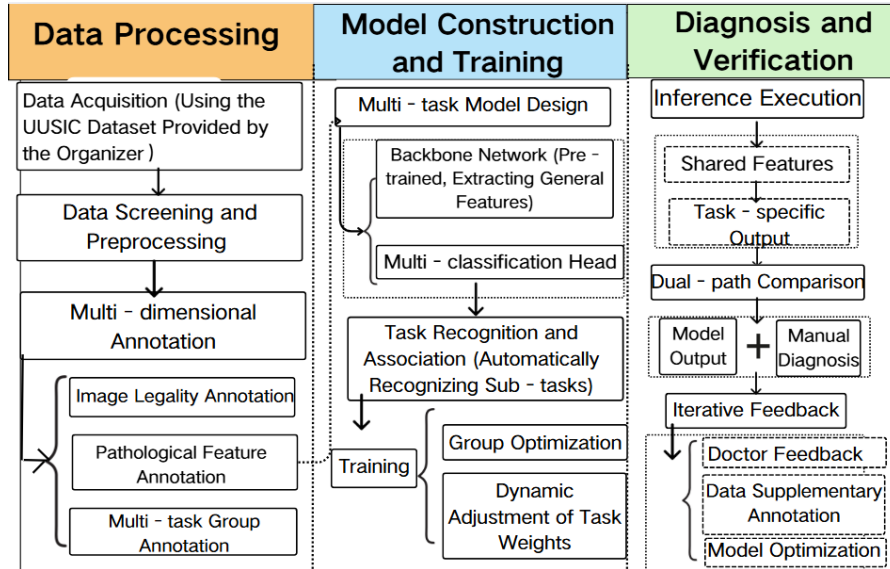


Fig. 3. Flowchart of the multi - task organ classification model: Improving EfficientNet into a multi - head architecture to exploit internal feature correlations across organ classification tasks, with modules for task recognition, multi - scale fusion, and efficiency enhancement. ut to segmentation output with atrous interpolation and transformer - related operations.

The core innovation of the model lies in its dynamic task adaptation ability, which realizes automatic task recognition by parsing the root file names of the input data. Specifically, the model can parse the organ identification in the root file name in real - time (such as "breast") and then activate the corresponding organ classification task module, effectively avoiding the model switching cost in multi - task scenarios and enabling a single model to adapt to the classification requirements of different organs and flexibly respond to diverse medical image analysis tasks.

On the top layer of the EfficientNet base network, we attach multiple independent classification heads to construct a multi - head structure and achieve multi - scale feature fusion. Different from the traditional single - head design, each classification head does not rely solely on high - level features but fuses the features of different network levels through skip connections. Low - level features retain details such as organ edges and textures to assist in lesion identification, while high - level features extract global semantics to support organ attribute judgment. Moreover, the output dimension of the classification head can be dynamically adjusted according to task requirements to specifically optimize the classification effect of specific types of organs. To adapt to the heterogeneity of ultrasound images, the model makes efficiency improvements in many aspects: optimizing the connection between the base model and the classification head to reduce the parameter scale while retaining the feature mapping ability; introducing an ultrasound data augmentation strategy and adjusting the learning rate scheduling mechanism; and transferring the feature extraction ability of the pre - trained EfficientNet to specific organ tasks through knowledge distillation technology^[7].

2.3 Loss Function^{[8][9][11]}

The core function of the loss function is to quantify the difference between the model's prediction results and the actual labels, and optimize the parameters through backpropagation to reduce the deviation between the model output and the actual values^[8]. In this model, we designed a hierarchical loss strategy based on the characteristics of multi - tasks^[9].

The segmentation model uses a combination strategy of Dice Loss + Cross Entropy Loss to optimize the parameters of the nnU - Net model, balance the similarity of the segmented region and the category classification loss, and improve the segmentation accuracy. Dice Loss focuses on the morphological matching degree of the segmented region. The Dice coefficient between the predicted mask and the real mask is calculated according to the following formula for optimization:

$$\zeta_{Dice} = 1 - \frac{2 \sum_{i=1}^N p_i g_i + \epsilon}{\sum_{i=1}^N p_i^2 + \sum_{i=1}^N g_i^2 + \epsilon}$$

where p_i is the pixel probability of the predicted mask, g_i is the real mask label, and ϵ is used to avoid a zero denominator. This loss function has stronger robustness to the "extremely unbalanced foreground - background" problem in multi - task medical image segmentation (such as the segmentation of small organs and small lesions), ensuring that the model can capture fine contours. To obtain a more accurate image, Cross Entropy Loss is used to supplement the global consistency constraint of category classification according to the following formula, enabling the model to learn the pixel - level category probability distribution and strengthen the semantic accuracy of the segmented region.

$$\zeta_{CE} = -\frac{1}{N} \sum_{i=1}^N g_i \log(p_i)$$

The weighted fusion of the two ($\zeta_{seg} = \alpha \zeta_{Dice} + (1 - \alpha) \zeta_{CE}$ where α is the balance coefficient) ensures both accurate segmentation morphology and consistent category annotation, adapting to the fine - grained segmentation requirements of nnUNet for multi - organ boundaries (such as mammary ducts and kidney contours).

In addition, the classification model uses a multi - task cross - entropy loss. Each classification head independently calculates the cross - entropy loss, and the total classification loss is the sum of the losses of the currently activated heads. The formula is as follows:

$$\zeta_{cls} = \sum_{k=1}^K \zeta_{CE,k} = \sum_{k=1}^K \frac{1}{N_k} \sum_{i=1}^{N_k} g_{i,k} \log(p_{i,k})$$

Where K is the number of activated classification heads, N_k is the sample size of the k-th head, and $g_{i,k}$ 、 $p_{i,k}$ are the real labels and predicted probabilities of the k-th head. This strategy allows each head to focus on learning the classification standards of the corresponding organs (such as breast BI - RADS classification and discrimination of liver lesion malignancy), adapting to the differences in clinical classification rules for different organs. At the same time, to avoid training imbalances between the segmentation and classification tasks and among the classification heads, a multi - task loss balance strategy is adopted, and a dynamic weight adjustment mechanism is introduced.

The total loss is defined as:

$$\zeta_{Total} = \omega_{seg} \zeta_{seg} + \omega_{cls} \zeta_{cls}$$

Where ω_{seg} 、 ω_{cls} are dynamic weights, initially set to 1. During training, they are adaptively adjusted according to the performance on the validation set. If the improvement of the Dice coefficient of the segmentation task is slower than the

accuracy of the classification task, then ω_{seg} ; otherwise, it is decreased. The weight update rules can be expressed as:

$$\omega_{seg}^{(t+1)} = \omega_{seg}^{(t)} \cdot \frac{Acc_{cls}^{(t)}}{Dice_{seg}^{(t)}}$$

$$\omega_{cls}^{(t+1)} = \omega_{cls}^{(t)} \cdot \frac{Dice_{seg}^{(t)}}{Acc_{cls}^{(t)}}$$

Where t is the number of iteration rounds, Acc_{cls} is the classification accuracy, and $Dice_{seg}$ is the segmentation Dice coefficient.

3 Experimental Results

3.1 Implementation Details

In the experimental implementation details, the data preparation stage uses the UUSIC dataset provided by the organizer, which is a combination of public and private ultrasound image datasets of different organs. For the nnUNet model of the segmentation task, it is processed by data augmentation techniques such as resampling, normalization, and random flipping, rotation, and scaling. Similar techniques are also used to enhance the data of the classification model to improve the diversity of the training data and the generalization ability of the model. During the training process, the segmentation model (nnUNet) and the classification model (improved multi - head EfficientNet) are first trained independently. The segmentation task trains nnUNet using the standard nnUNet training program, and the classification task trains the classification model based on EfficientNet. Then, they enter the joint training stage for fine - tuning, combining the two models and considering the multi - task loss function for joint optimization. The training is carried out on a machine with GPU support, and the training parameters such as batch size, learning rate, and number of iterations are adjusted according to the performance of the validation set^{[10][11][12]}.

3.2 Experimental Results

To verify the effectiveness of the method, experiments were carried out to evaluate two aspects: one is the effectiveness of the dual - branch multi - task network compared to a single - branch complete independent network; the other is the effectiveness of the multi - task network compared to commonly used current segmentation and classification networks. In this research model, the classification branch of the Transformer encoder uses DeiT, and the convolutional branch absorbs the encoding - decoding idea of the U - Net architecture, fusing the feature maps of different levels of the encoder and outputting the final segmentation map through the skip - connection decoder.

In the segmentation task, the segmentation model based on nnUNet demonstrates favorable performance in terms of the Dice Similarity Coefficient (DSC) for different organs on the validation set. For instance, the average DSC of breast tissue segmentation (encompassing all breast-related categories) is 0.853. Notably, the DSC values for other organs are also impressive, such as 0.847 for cardiac segmentation, 0.948 for fetal head segmentation, 0.890 for kidney segmentation, and 0.721 for thyroid segmentation. These results are comparable to those of other advanced segmentation methods on similar datasets, confirming the effectiveness of nnUNet in multi-organ segmentation tasks. In the classification task, the multi-head classification model also yields decent outcomes, with varying classification accuracies across different organs. The validation set accuracy of breast lesion classification is 0.591, while that of appendix classification is 0.507 and liver classification is 0.529. Although its accuracy might be lower than that of dedicated single-task models, its advantage lies in being able to handle multiple tasks within a more lightweight framework.

4 Discussion and Conclusion

4.1 Discussion

This research shows significant advantages and certain limitations in the design and practice of multi - task medical image analysis models. In terms of advantages, a lightweight deployment strategy that integrates classification and segmentation in a single model is adopted, abandoning the multi - model splicing mode, effectively reducing system complexity, adapting to resource - constrained scenarios such as mobile ultrasound and primary - level medical care, and reducing deployment and operation costs. The innovative "root file name - task head" dynamic adaptation mechanism breaks through the traditional model - solidified mode. When new organ tasks are added, only data needs to be supplemented and classification heads need to be expanded, enabling flexible responses to new medical research requirements. With the collaboration of classic models, the segmentation reuses the accuracy of nnUNet, and the classification improves the multi - head structure based on EfficientNet, balancing feature reuse and task specificity, conforming to the clinical diagnosis process, inheriting the advantages of classic models, and exploring the value of multi - tasks.

However, the model also has limitations. The competition between "general - purpose and task - specific features" in multi - task learning is significant. Facing highly heterogeneous tasks such as the classification of small lesions and rare diseases, shared features are likely to dilute specific information, resulting in lower accuracy than single - task models. There is an urgent need to explore feature decoupling

mechanisms. The current "validation - set index - adjusted weight" strategy has issues of hysteresis and local optimality. During long - cycle training or when the task distribution changes, it is difficult to match the task difficulty in real - time, leading to problems such as loss of segmentation details and false positives in classification. It is necessary to introduce intelligent control means such as reinforcement - learning - based dynamic weights to optimize the model performance^[13].

4.2 Conclusion

This model demonstrates the feasibility of using a combined model for multi - organ classification and segmentation tasks in ultrasound images. By using nnUNet for segmentation and an improved multi - head EfficientNet for classification, a balance between performance and efficiency is achieved. Although there are limitations in the accuracy of some tasks and there is a need to further optimize the loss function, the overall method shows great potential in the future development of the field of medical ultrasound image analysis. In future work, we will further improve the model architecture, loss function, and training strategy. We believe that our model can achieve better performance and make more important contributions to the clinical application of ultrasound image analysis.

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